Solving System of Equations

Solve each of the following systems by replacing each system by an equivalent system.

a) $\begin{aligned} 3x+2y-4z&=-7\\ 2x-y+5z&=13\\ x-6y+2z&=11 \end{aligned}$

b) $\begin{aligned} x+3y-z&=3\\ 2x-4y+z&=4\\ 3x+2y+2z&=-4 \end{aligned}$

c) $\begin{aligned} 2x+7y-z&=28\\ 3x-y+2z&=-10\\ 4x+y+5z&=-18 \end{aligned}$

d) $\begin{aligned} x+2y-z&=-5\\ 3x-y-2z&=3\\ 4x+3y+3z&=18 \end{aligned}$

e) $\begin{aligned} 5x+y-z&=-2\\ 2x+2y-3z&=3\\ x+3y-4z&=7 \end{aligned}$

ANSWERS:

a) $(x,y,z)=(1,-1,2)$ b) $(x,y,z)=(2,-1,-4)$ c) $(x,y,z)=(1,3,-5)$

d) $(x,y,z)=(3,-2,4)$ e) $(x,y,z)=(-1,4,1)$